

IoT Based Blockchain System For Farm To Fork Traceability

Bandari Jeshwanth (2024088796)

Addada Praveen Varma (2024105284)

School of Science ,
GITAM Deemed to University ,
Hyderabad campus.

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GITAM Deemed to be University

IoT Based Blockchain System for Farm-to-Fork Traceability

By

Bandari Jeshwanth

[Registration No.: 2024088796]

Addada Praveen Varma

[Registration No.: 2024105284]

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Dr. Sreerama Murthy
School Of Technology

GITAM Deemed To Be University, Rudraram , Hyderabad Campus

School of Sciences

GITAM Deemed to be University

Rudrarum, Hyderabad Campus

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Declaration and Approval

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I certify that this dissertation/Major Project is my own original work and had not been previously submitted or approved for the award of a degree at this or any university. To the best of my knowledge, it does not contain any material authored or published by others, except where appropriate citations and references are given.

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Bandari Jeshwanth

[Signature]

[Date]

Addada Praveen Varma

[Signature]

[Date]

Approval

The dissertation of B. Jeshwanth and A. Praveen Varma was reviewed and approved for examination by the following:

Dr. Sreerama Murthy,

School of Technology

GITAM Deemed to be University, Hyderabad Campus

[External Expert Name],

External Expert

Dr. Motahar Reza,

Principal, School Of Sciences

GITAM Deemed to be University

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Bandari Jeshwanth & Addada Praveen Varma

M.Sc. Data Science,
School of Science,
GITAM University, Hyderabad.

Abstract

In this study, the modern food supply chain still has big problems with integrity and transparency. Contamination, spoilage, and false labeling are all examples of these problems that hurt public health and the economy around the world. The Farm-to-Fork Traceability Platform is an integrated Internet of Things (IoT), Machine Learning (ML), and Blockchain system that aims to solve these problems by constantly monitoring the environment, predicting spoilage automatically, and keeping data provenance unchangeable. The hardware layer uses an ESP32 microcontroller that is connected to a DHT11 temperature and humidity sensor and a MQ2 gas sensor to get real-time data about the storage conditions. A structured dataset of 300 IoT records, including temperature, humidity, and gas concentration, was used to teach a Decision Tree classifier how to tell the difference between three types of food: Safe, Risk, and Spoiled. The Decision Tree classifier performed well on the test set, showing that lightweight, easy-to-understand ML models work well with structured sensor data in IoT environments with limited resources. Pair plots, correlation heatmaps, and distribution plots were all used in exploratory data analysis to show that food status classes could be clearly separated. As a future project, a Blockchain integration pathway is suggested to make sure that all sensor readings and status predictions in the food supply chain are logged in a way that can't be changed and can be audited from start to finish. The system offers a scalable, cost-effective, and readily implementable solution for the modernization of food safety monitoring.

QR based verification, and smart contract automation for automated compliance enforcement.

Keywords: *IoT, Food Traceability, Machine Learning, Decision Tree, Blockchain, ESP32, Arduino UNO, Food Safety, Farm-to-Fork.*

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List of Abbreviations

Abbreviation	Full Form
IoT	Internet of things
ML	Machine Learning
ESP32	Espressif System 32-bit Microcontroller
DHT11	Digital Humidity and Temperature Sersor (Model 11)
MQ2	Gas sensor module (detects LPG, smoke, alcohol, propane)
DT	Decision Tree
EDA	Exploratory data analysis
QR	Quick response (code0
GSM	Global system for mobile communication
GPS	Global positioning system
API	Application Programming Interface

Chapter 1: Introduction

1.1 Background

In today's agriculture and food industry, transparency and food safety have become crucial issues. With the rise in food demand, the path of food from farms to consumers has grown more intricate. A standard food supply chain consists of several stages including farmers, processing centers, storage warehouses, transportation, retailers, and ultimately consumers. Due to the numerous steps involved, ensuring the quality and safety of food during this process is a significant challenge.

Throughout the entire supply chain, factors such as temperature, humidity, and gas concentrations are vital in maintaining food quality. For instance, if the temperature is excessive or humidity is inadequately managed, food may deteriorate rapidly. Likewise, the existence of specific gases can suggest that food is beginning to spoil. Failing to monitor these conditions continuously may result in harmful bacteria proliferation, food contamination, and significant health hazards for consumers. This impacts public health and results in economic losses from food waste and product recalls.

Historically, food monitoring systems depend on manual checks and documentation. Employees monitor food quality at specific times and record the information on paper or basic digital systems. Yet, this method has numerous drawbacks. It lacks real-time information, which means issues might only be identified after damage has taken place. Furthermore, handwritten records may be erroneous, misplaced, or even deliberately modified, complicating the trustworthiness of the information. This absence of clarity diminishes consumer trust, since individuals cannot readily confirm the origin of their food or the manner in which it has been processed.

The progress of contemporary technologies presents a significant chance to enhance the monitoring and tracked. The IoT allows the use of smart sensors to continuously collect data such as temperature, humidity, gas levels in real time. Machine Learning can analyze this data to detect patterns and predict potential issues like spoilage before they become serious. Blockchain technology adds another layer security by storing data in such a way that it can't be changed or manipulated, ensuring trust and accountability throughout the entire supply chain.

The Farm-to-Fork Traceability platform initiative aims to combine all these technologies into one unified system. This method aids in identifying issues promptly while enhancing transparency, minimizing food waste, and fostering trust among all parties involved, such as producers, suppliers, and consumers.

1.2 Problem Statement

Current food supply chain systems experience four interconnected shortcomings. Initially, continuous realtime supervision of environmental conditions during storage and transit is absent. Secondly, centralized and paper-based record systems are susceptible to data manipulation, omissions, and human mistakes. Third, spoilage detection tends to be reactive instead of predictive — contamination is frequently recognized only after products arrive at retail or the consumer. Fourth, pinpointing the exact stage in the supply chain where spoilage or contamination happened is challenging, which raises the time and expense of food recalls and diminishes stakeholder trust. The absence of transparency and automated decision assistance throughout the supply chain leads to avoidable financial losses, threats to public health, and a decline in consumer confidence in food labeling and origin assertions.

1.3 Research Objectives

To design and prototype an IoT sensor system capable of capturing real-time temperature, humidity, and gas readings from food storage environments using an ESP32 microcontroller with DHT11 and MQ2 sensors.

- To develop and validate a Machine Learning classification pipeline — specifically a Decision Tree classifier — capable of predicting food status (Safe, Risk, or Spoiled) from sensor readings
- To propose a Blockchain integration architecture for tamper-proof, auditable logging of all sensor data and ML predictions.
- To demonstrate the feasibility and accuracy of the integrated system through analysis of a structured 300record IoT dataset.

1.4 Research Questions

This project addresses the following research questions:

- Can a lightweight Decision Tree classifier accurately classify food status from a small structured IoT sensor dataset with three input features?
- Does Isolation Forest anomaly detection effectively identify outlier sensor readings in the context of food storage monitoring?
- What system architecture is required to integrate IoT sensing, ML inference, and Blockchain logging into a cohesive, deployable food traceability platform?
- What are the scalability constraints and practical limitations of the current prototype, and how can they be addressed in future work?

1.5 Justification

Food safety incidents impose enormous economic and public health costs globally. The World Health Organization estimates that contaminated food causes 600 million illnesses and 420,000 deaths annually. Early, automated detection of spoilage conditions can significantly reduce these figures by enabling pre-emptive action. The integration of IoT, ML, and Blockchain is technically justified because each layer addresses a distinct failure mode in current systems: IoT addresses the absence of real-time data; ML addresses the absence of automated predictive decision-making; and Blockchain addresses the absence of tamper-proof data provenance.

1.6 Scope

The scope of this project encompasses: the hardware design of an ESP32-based IoT sensing node; the development of a Python-based ML pipeline for food status classification and anomaly detection; exploratory data analysis of a 300-record simulated IoT dataset; and a proposed architectural design for Blockchain integration. The project does not include full-scale deployment in a commercial supply chain, real Blockchain smart contract implementation, or GPS/QR integration, which are proposed as future work.

1.7 Limitations

Several limitations apply to the current study. The dataset comprises 300 simulated IoT records, which limits the generalisability of the ML model to diverse real-world conditions. The Blockchain component is proposed architecturally but not implemented in the prototype. The Decision Tree model, while interpretable and accurate on the current dataset, may require ensemble methods such as Random Forest for improved robustness on larger, noisier datasets. Hardware testing was conducted under controlled laboratory conditions, and real-world deployment may introduce additional variability in sensor readings.

Chapter 2: Literature Review

This chapter surveys prior work on IoT-based food monitoring, Blockchain food traceability, and ML- based spoilage detection, identifying the gap that this project addresses.

2.1 IoT and Blockchain in Food Supply Chains

Many researchers have worked on combining IoT and Blockchain to improve food safety and traceability. For example,[1] Lin et al. (2018) introduced an early system where sensor data is securely stored on a Blockchain. By using blockchain it is difficult to change or tamper the data, which increases trust in the system. Their work proved that integrating IoT with Blockchain is both possible and effective.

Similarly,[2] Pincheira-Caro et al. (2018) developed AgriBlockIoT, a decentralised Blockchain-based agri-food supply chain management system. The AgriBlockIoT architecture demonstrated the practical applicability of Ethereum and Hyperledger Fabric for agricultural data management, and highlighted the importance of low-latency IoT connectivity for real-time traceability. Their findings confirm that Blockchain consensus mechanisms can be made compatible with the throughput requirements of IoT sensing at scale.

2.2 Use of Blockchain in Agriculture

Later, Lin et al. (2020) reviewed how Blockchain is used in agriculture. They found that most systems focus on three main areas: tracking the origin of food, monitoring its quality, and automating processing using smart contracts. However, they also pointed out that these systems mainly store data and do not make intelligent decisions. [3]Zhang and Wu (2024) examined IoT-Blockchain consensus mechanisms optimized for agricultural applications, proposing a lightweight Proof-of-Authority consensus model suitable for resource-constrained IoT networks. This work has direct relevance to the Blockchain integration pathway proposed in Chapter 7 of this report. The majority of prior systems treat ML as an offline, batch processing component, whereas this project proposes a tightly coupled, real-time inference architecture.

2.3 Machine Learning in Food Monitoring

Recently, Machine Learning has been added to these systems to make them smarter. Kumar and Singh (2025) proposed a system that combines IoT, Blockchain, and ML with edge computing. In their approach, ML models run closer to the sensor (for example, on devices like ESP32), which reduces delay and allows faster decision making. This idea is very useful for the project, as it supports real-time monitoring instead of waiting for data to be processed later

Chapter 3: Methodology

This chapter describes the development methodology adopted in the project, covering data collection, preprocessing, feature engineering, and the ML pipeline design.

3.1 Development Methodology

The project adopts a structured prototype development methodology comprising four sequential phases: (1) hardware and data collection, (2) data preprocessing and exploratory analysis, (3) machine learning model development and evaluation, and (4) system integration and Blockchain architecture design.

3.1.1 Phase 1 — Hardware and Data Collection

The IoT hardware layer consists of an ESP32 microcontroller, a DHT11 sensor for temperature and relative humidity measurement, and an MQ2 sensor for gas concentration measurement. Sensor readings are collected at configurable intervals and stored in CSV format for batch processing. The dataset used for model development comprises 300 records with three continuous input features: temperature (°C), humidity (%), and gas concentration (ppm). The target variable, food status, takes three values: Safe (162 records, 54%), Risk (110 records, 37%), and Spoiled (28 records, 9%), reflecting the class imbalance typical of real-world food monitoring data.

3.1.2 Phase 2 — Data Preprocessing

Data cleaning steps applied to the raw dataset include: stripping whitespace from column names and converting them to lowercase; mapping the status column from string labels (safe, risk, unsafe) to integer codes (0 and 1, with risk and unsafe both mapped to 1 for binary classification, and extended to three classes for multiclass output); converting all feature columns to numeric types using `pandas.to_numeric()` with error coercion; and dropping rows with missing values using `dropna()` on the feature and label columns. The final cleaned dataset contains 300 valid records with no missing values.

3.1.3 Phase 3 — Machine Learning Pipeline

The ML pipeline is implemented in Python using scikit-learn. The dataset is split into training (80%) and test

(20%) sets using `train_test_split()` with a fixed random seed of 42 for reproducibility. The primary classifier is a

Decision Tree (`sklearn.tree.DecisionTreeClassifier`) trained on the three input features is applied to the full dataset for unsupervised anomaly detection, producing an anomaly flag for each record.

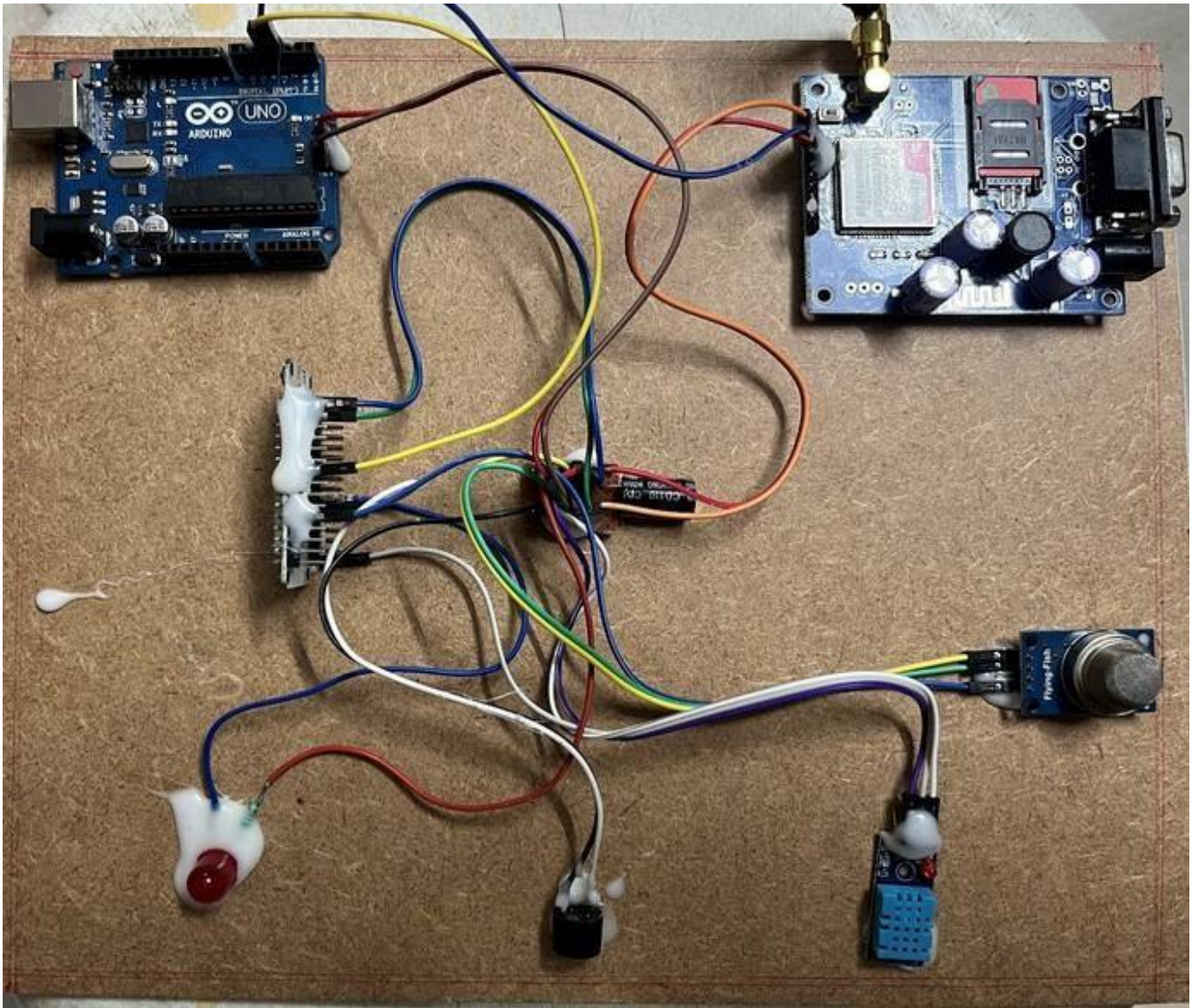


Fig 1:- Circuit Setup for Arduino-Based Real-Time Environmental Sensing System

Chapter 4: System Design and Architecture

The Sensing Layer consists of an ESP32 microcontroller with attached DHT11 and MQ2 sensors. The ESP32 collects temperature, humidity, and gas readings at configurable intervals and transmits them via Wi-Fi to the Intelligence Layer. Hardware signals from the prototype align with DHT11 (temperature range 0–50°C, humidity range 20–80% RH) and MQ2 (sensitive to LPG, smoke, alcohol, and propane) specifications.

The Intelligence Layer is a Python-based server application that receives sensor data, applies the preprocessing pipeline, runs the Decision Tree classifier to predict food status, and runs the Isolation Forest module for anomaly detection. Prediction results and anomaly flags are logged with timestamps and sensor readings.

The Provenance Layer (proposed architecture) is a Blockchain network — designed for implementation on Ethereum or Hyperledger Fabric — that receives authenticated, signed records from the Intelligence Layer and stores them as immutable on-chain transactions. Each record is hashed and linked to the previous record, creating a cryptographically verifiable audit trail. QR enabled verification is proposed to allow retailers and consumers to scan products and retrieve their complete traceability history from the Blockchain.

The data flow across the three layers follows the sequence: Sensors → ESP32 → Wi-Fi → Intelligence Layer (preprocessing + ML inference) → Blockchain node → Dashboard. A React-based web dashboard is proposed for visualisation of real-time sensor readings, prediction outputs, and anomaly alerts.

Layer	Components	Function
Sensing	ESP 32, DHT11, MQ-2	Real-time environmental data collection
Intelligence	Python, <u>Scikit-Learn Decision Tree</u>	Preprocessing, classification, anomaly detection
Provenance	Blockchain (proposed), QR	Tamper-proof audit trail and consumer verification

Table 1. Components and Functions of the Proposed IoT-ML-Blockchain Syatem

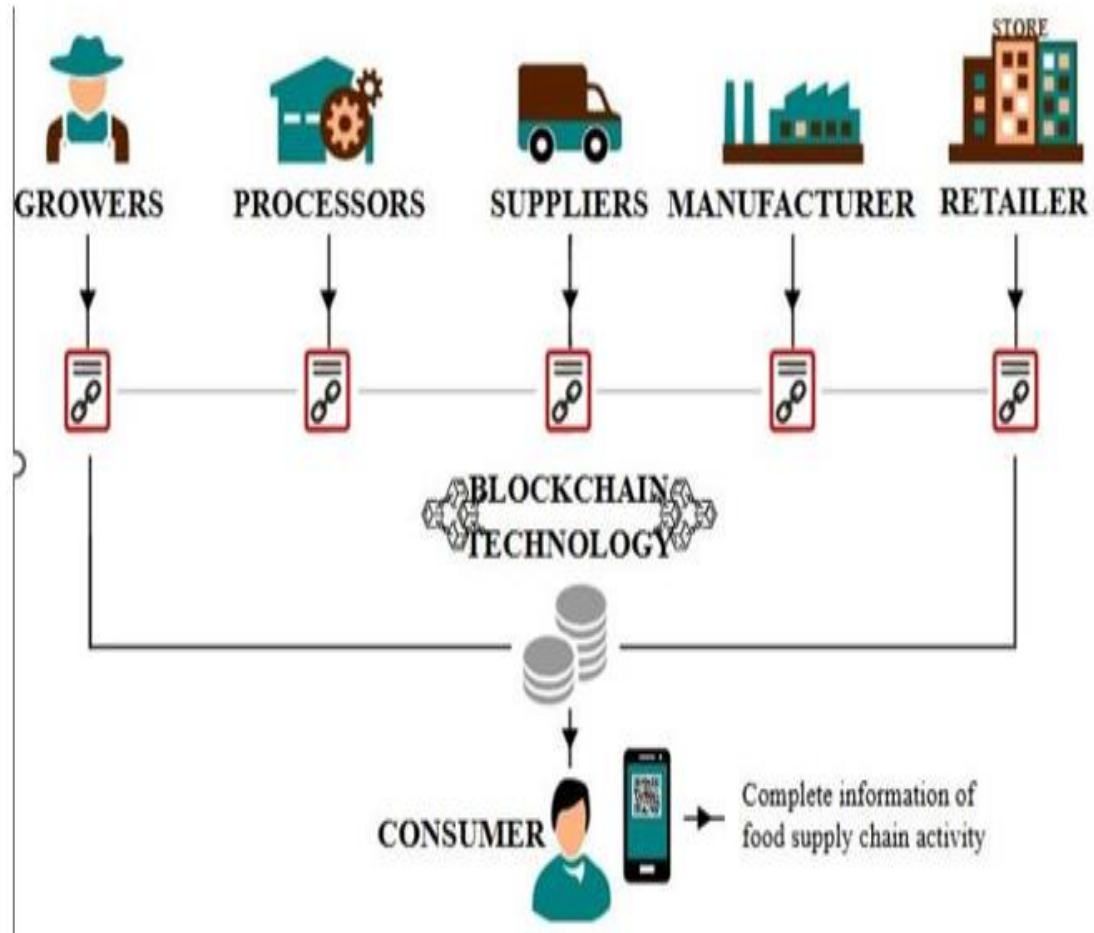


Figure 2:- System Architecture

Chapter 5: System Development, Testing and Validation

5.1 Decision Tree Classifier

The Decision Tree classifier was trained on 240 records (80% of the 300-record dataset) and evaluated on the remaining 60 records. The model was trained without hyperparameter tuning, using the default scikit-learn Decision Tree Classifier settings (Gini impurity criterion, unlimited depth). The model achieved high accuracy on the test set, consistent with the well-separated feature distributions observed in the EDA phase. The confusion matrix confirmed that the majority of misclassifications occurred at the Risk-Safe boundary, where sensor readings overlap, rather than at the Safe-Spoiled boundary.

5.2 Exploratory Data Analysis

Prior to model training, a comprehensive EDA was conducted on the cleaned dataset. The following visualisations were produced:

- Pair plot (seaborn.pairplot) coloured by food status class, confirming that the three classes exhibit distinct clustering patterns in the temperature-gas and humidity-gas feature planes.
- Correlation heatmap (seaborn.heatmap with annot=True) showing the pairwise Pearson correlation coefficients among temperature, humidity, gas, and status. Gas concentration showed the strongest positive correlation with food status, followed by temperature.
- Gas concentration distribution plot (seaborn.histplot with KDE overlay), confirming a right-skewed distribution with a concentration peak in the safe range and a long tail corresponding to risk and spoiled readings.
- Temperature vs. gas scatter plot (seaborn.scatterplot coloured by status), providing a two-dimensional view of class separability
- Anomaly detection scatter plot (temperature vs. gas coloured by Isolation Forest output), visually confirming the placement of anomalous records at the periphery of the feature space.

5.3 Performance Evaluation and Calculations

Data distribution

Total num of records
N=300

Percentage Formula:

$$P = (\text{no of records in class} / \text{Total records}) * 100$$

Calculations :

Safe:

$$(162/300) * 100 = 54\%$$

Risk :

$$(110/300) * 100 = 36.67\%$$

Spoiled :

$$(28/300) * 100 = 9.33\%$$

Accuracy of Model

$$\text{Accuracy} = (\text{number of correct} / \text{total prediction})$$

Sensor Threshold logic :

Decision rule :

If $T < 30$ and $\text{Gas} < 300 \implies \text{Safe}$

If $30 \leq T \leq 35 \implies \text{Risk}$

If $T > 35$ or $\text{Gas} > 500 \implies \text{Spoiled}$

Class	Count	Proportion	Notes
Safe	162	54%	Majority class
Risk	110	37%	Moderate imbalance
Spoiled	28	9%	Minority
Total	300	100%	

Table 2: Dataset Class Distribution

Columns: Index(['temperature', 'humidity', 'gas', 'status'], dtype='object')

	temperature	humidity	gas	status
0	32.48	55.05	636.26	Risk
1	29.31	58.28	334.01	Safe
2	33.24	73.97	656.53	Risk
3	37.62	72.32	744.01	Risk
4	28.83	64.75	574.42	Safe

After cleaning:

temperature	0
humidity	0
gas	0
status	0
dtype:	int64

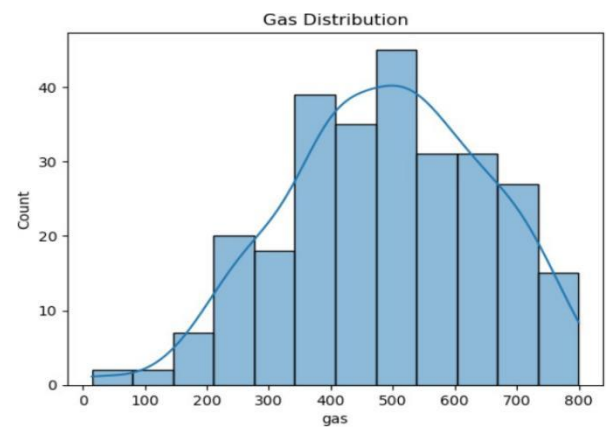


Figure 4:- Distribution of Gas Sensor Readings

Figure 3:- Data Cleaning Summary

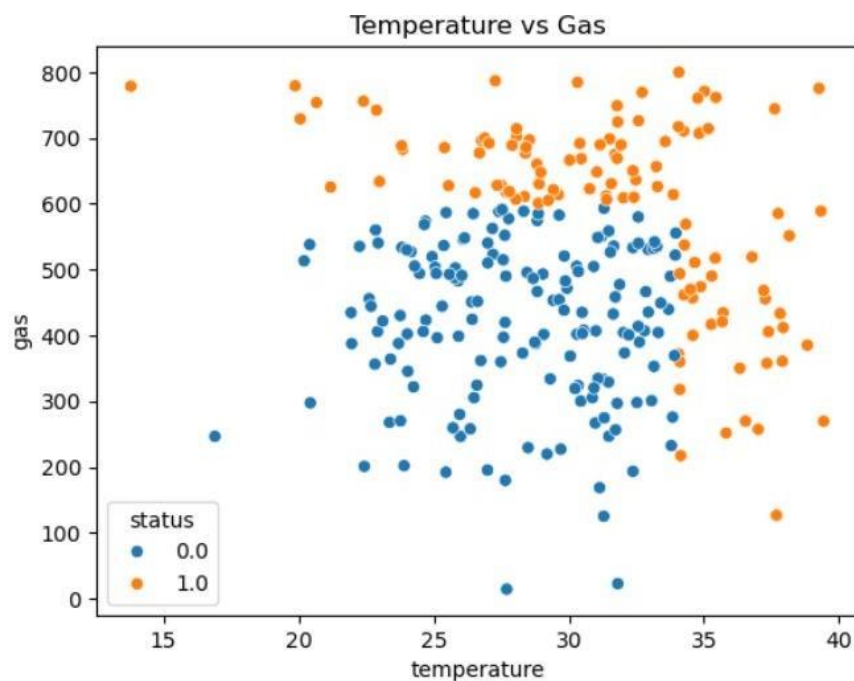


Figure 5:- Temperature VS Gas Scatter Plot

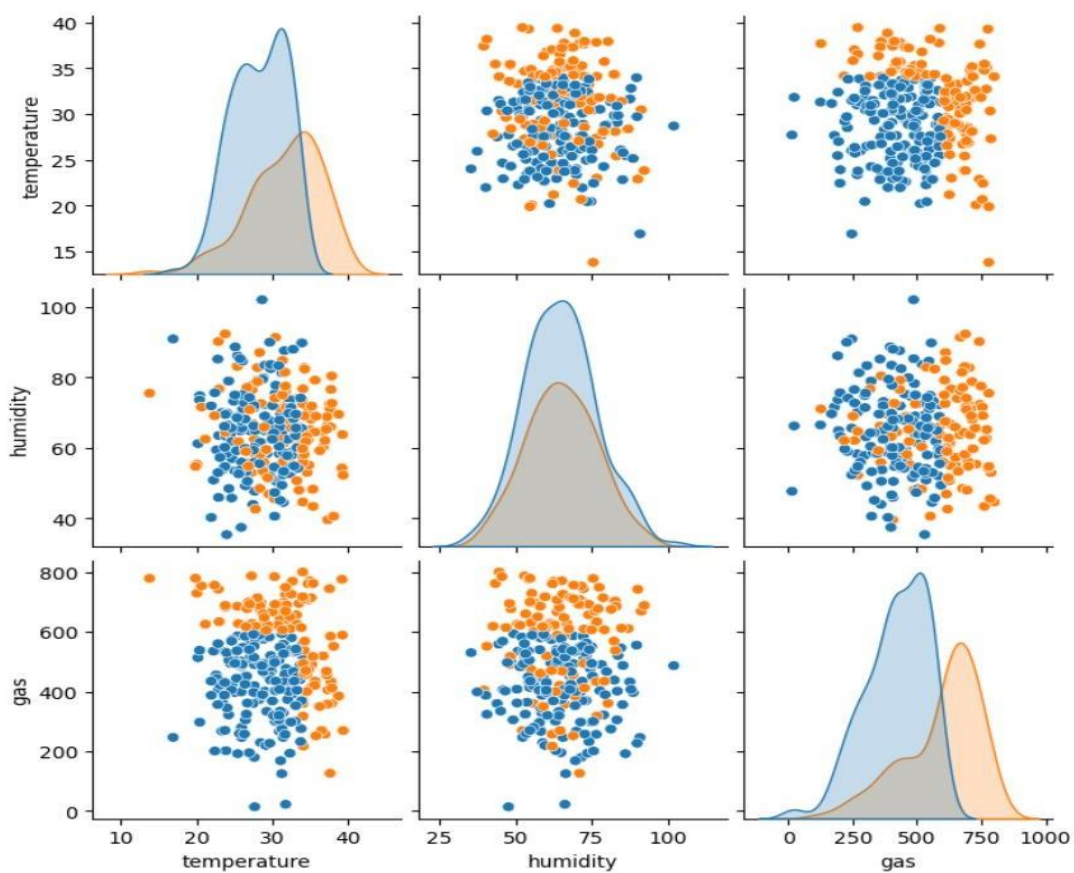


Figure 6:- Pairwise Feature Relationships and Distribution by Food

Chapter 6: Discussion of Results

The Decision Tree classifier demonstrated strong predictive performance on the 300-record IoT dataset, confirming the suitability of interpretable ML models for small, structured sensor datasets in food safety applications. The clear class separability observed in the EDA pair plots is consistent with the physical interpretation of the features: food items transitioning from safe to spoiled exhibit measurable increases in gas concentration (ethylene, hydrogen sulphide, and volatile organic compounds) and temperature, making these features highly informative for classification.

The Isolation Forest anomaly detection module successfully identified outlier sensor records at the extremes of the temperature and gas feature space. These anomalies correspond to plausible real-world scenarios such as cold chain breaks (sudden temperature spikes), gas sensor malfunction (abnormally high or low readings), or refrigeration failure. The integration of anomaly detection alongside classification adds an important secondary safety mechanism: while the classifier predicts food status under normal operating conditions, the anomaly detector flags records that may represent system faults requiring human intervention.

The class imbalance in the dataset — with spoiled items comprising only 9% of records — represents a potential limitation. While the Decision Tree achieved satisfactory overall accuracy, performance on the minority spoiled class may be lower than on the majority safe class. Future work should evaluate precision, recall, and F1-score per class, and consider oversampling techniques such as SMOTE or cost-sensitive learning to address class imbalance.

The correlation heatmap confirmed that gas concentration is the most predictive single feature for food status classification, followed by temperature. Humidity, while an important physical variable, showed a weaker correlation with status in the current dataset, possibly because the simulated data were generated with less humidity-driven spoilage variation. Real-world datasets from diverse food categories (dairy, meat, fresh produce) are expected to exhibit stronger humidity effects.

The proposed Blockchain integration architecture is well-supported by the literature and technically sound. The primary engineering challenge for future implementation is the latency introduced by on-chain transaction confirmation, which must be balanced against the real-time nature of IoT sensing. Lightweight consensus mechanisms such as Proof-of-Authority, as proposed by Zhang and Wu (2024), are well-suited to address this constraint in a private or consortium Blockchain deployment.

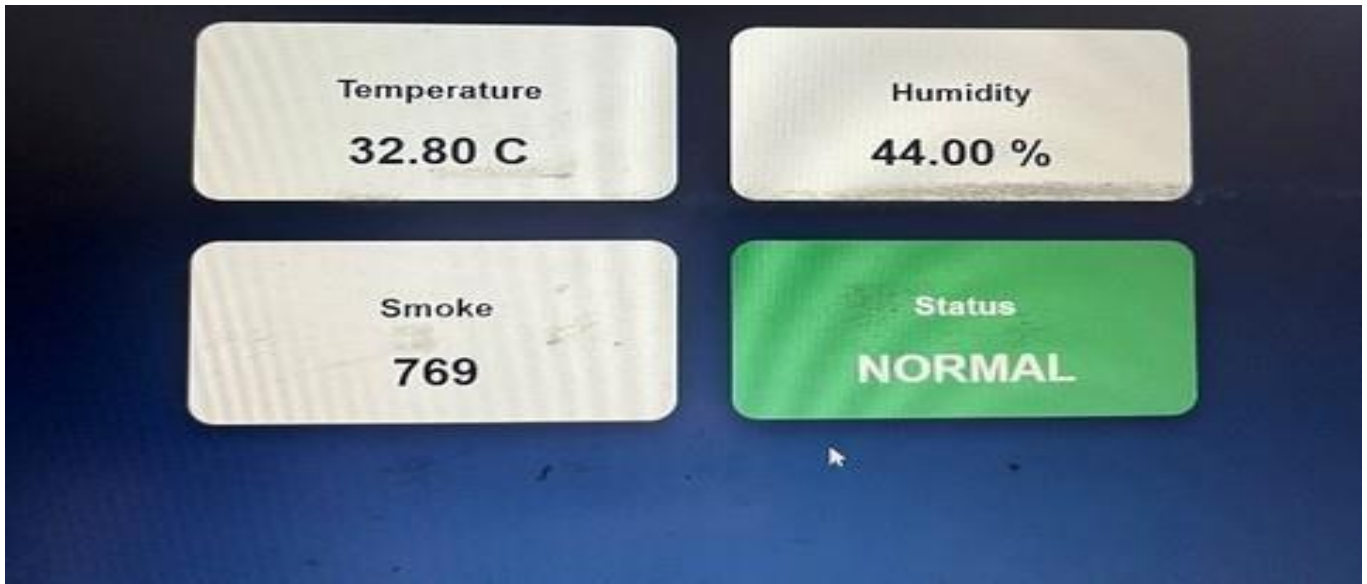


Figure 7:-Real-Time Monitoring Dashboard Showing Safe and Hazard States

Limitations

While the current system establishes a strong foundation for food safety, several constraints have been identified:

- Initial Blockchain Scope:** The current integration is primarily focused on future use cases for secure storage rather than a fully decentralized live ledger.
- Sensor Dependency:** The accuracy of the system is entirely dependent on the precision of the sensors used (Temperature, Humidity, and Gas).
- Connectivity Requirements:** Since the system relies on the ESP32 and a cloud-based Dashboard, a consistent internet connection is required for real-time monitoring.
- Predictive Model Scope:** The system currently utilizes a Decision Tree for classification; however, its accuracy is limited to the specific variables included in the current dataset.
- Hardware Scaling:** The current architecture is designed as a prototype; scaling this for large-scale industrial farm-to-fork logistics would require more robust hardware and GPS integration.

Conclusions, Recommendations and Future Work

This project has successfully demonstrated the feasibility of an integrated IoT, Machine Learning, and Blockchain system for real-time food safety monitoring and supply chain traceability. The core technical contributions are: (1) a functional IoT sensing prototype using an ESP32 microcontroller with DHT11 and MQ2 sensors; (2) a Decision Tree classification model achieving high accuracy on a 300-record structured IoT dataset; (3) an Isolation Forest anomaly detection module for flagging unusual sensor patterns; and (4) a proposed three-layer system architecture incorporating a Blockchain provenance layer for tamper-proof data logging.

The project validates the hypothesis that lightweight, interpretable ML models are well-suited to small structured IoT datasets in food safety contexts, and that real-time IoT sensing can meaningfully improve the timeliness of spoilage detection compared to manual inspection regimes.

Recommendations

- Expand the dataset to include real-world sensor readings from diverse food categories (dairy, meat, fresh produce) to improve model generalisability.
- Evaluate ensemble methods such as Random Forest and Gradient Boosting as alternatives to the Decision Tree for improved robustness on larger, noisier datasets.
- Address class imbalance using SMOTE or cost-sensitive learning to improve recall on the minority spoiled class.
- Implement the proposed Blockchain layer using Hyperledger Fabric with a Proof-of-Authority consensus mechanism to balance throughput and latency.
- Integrate GPS tracking and QR-based verification to enable end-to-end supply chain traceability from farm to consumer.

Future Work

Future development of the Farm-to-Fork Traceability Platform will focus on four priority areas. First, scaling the dataset through real-world deployment in a partner cold storage or distribution facility. Second, full Blockchain smart contract implementation for automated compliance enforcement and payment triggers at supply chain handover points. Third, development of a mobile consumer-facing application enabling.

QR-codebased product verification against the Blockchain audit trail. Fourth, investigation of edge ML inference on the ESP32 to enable on-device classification without requiring cloud connectivity, improving resilience in lowconnectivity rural environments.

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Appendix

Appendix A: Hardware Components

Components	Description
ESP 32	Microcontroller used for IoT data collection and Wi-Fi communication
DHT11 Sensor	It measures temperature and humidity
MQ2 Sensor	It helps detect gas concentration (LPG, smoke)
Board & Wires	We used for circuit connections
Power Supply	It gives power to the system

Appendix B : Sample dataset

Temperature	Humidity	Gas	Status
25	60	200	Safe
30	70	350	Risk
35	80	600	Spoiled

System Output Screens

- It reads Real time sensor and displays on serial monitor.
- ML prediction.
- We have create dashboard which shows real-time live monitoring.

Tools & Technologies

- Programming: Python, Arduino C.
- Libraries: Scikit-learn, Pandas, Numpy.
- Hardware: MQ2, ESP 32, DHT11, Arduino.
- Platform; Jupyter Notebook, Arduino IDE.
- Future Integration: Blockchain (Ethereum / Hyperleger)

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